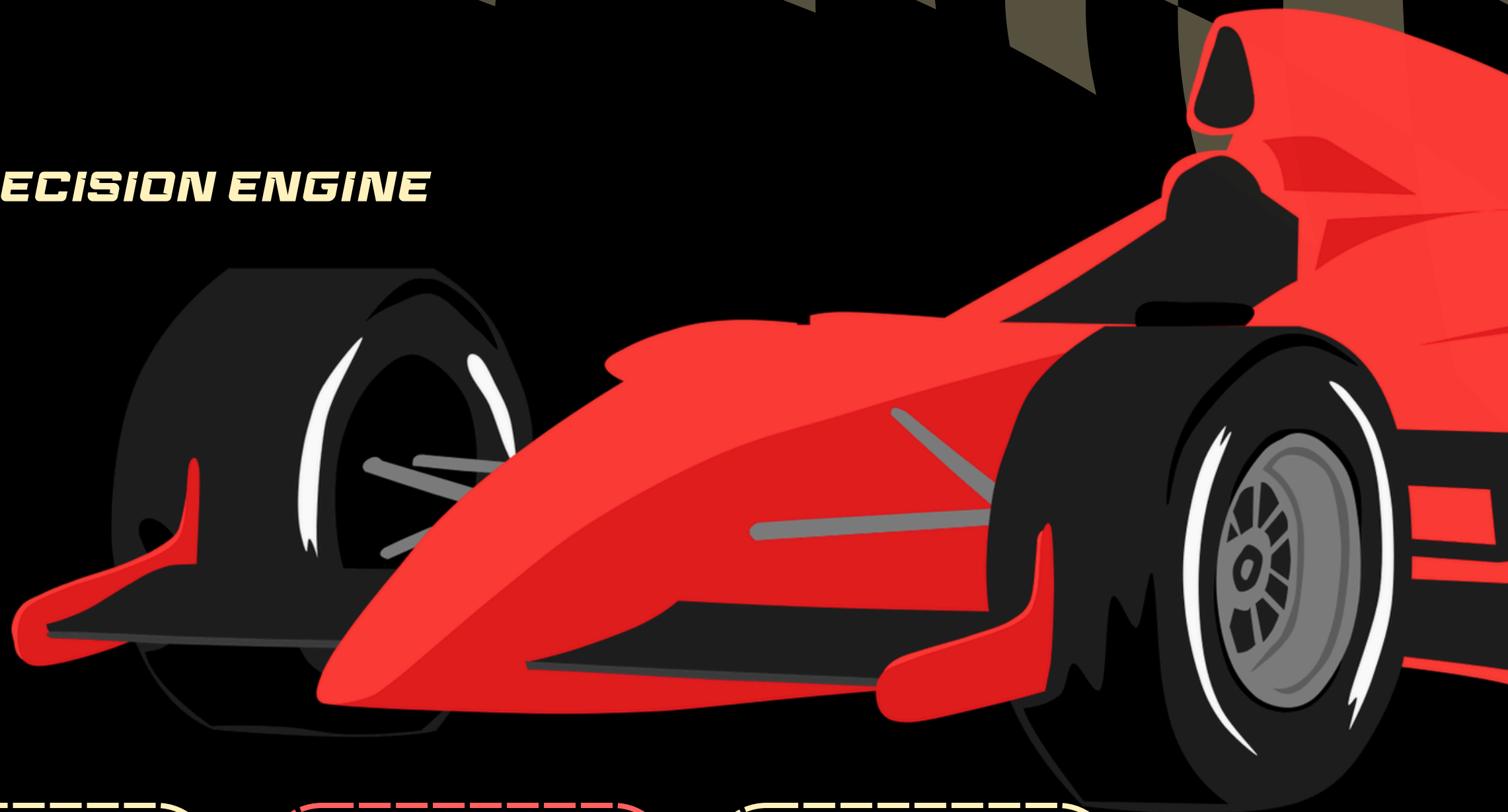


APEX

AI MOTORSPORT INTELLIGENCE
A REAL-TIME ENERGY & OVERTAKE DECISION ENGINE

BALANCING AGGRESSIVE TACTICAL
RACING AGAINST **LONG-TERM**
BATTERY, THERMAL, AND
REGULATORY LIMITS, WITH
MATHEMATICALLY GUARANTEED
SAFETY.



4

CONTROL
TIERS

200

HZ SAFETY
LOOP

<5

MS WORST-CASE
SAFETY RESPONSE

100%

HARD
CONSTRAINT
ENFORCEMENT

THE PROBLEM

01

ENERGY + POSITION UNDER UNCERTAINTY

F1 racing is not just about driving fast; it is an energy-and-position problem where the car knows its own state but cannot directly know rivals' battery, tyre life, or intent.



02

THE REAL OVERTAKING QUESTION

The goal is not simply to ask “Can we overtake now?” but whether spending energy on the manoeuvre improves the eventual race result.



03

WHY GREEDY DECISIONS FAIL

An overtake that is immediately lost, attacking without defence energy, or harvesting before a high-value straight can all lead to worse race outcomes.



04

BEYOND OVERTAKES: RACE OUTCOME

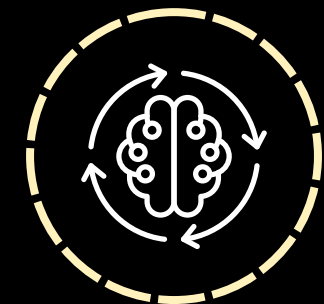
The system optimises for winning first, finishing second, and classified position third—rather than simply maximising the number of overtakes.



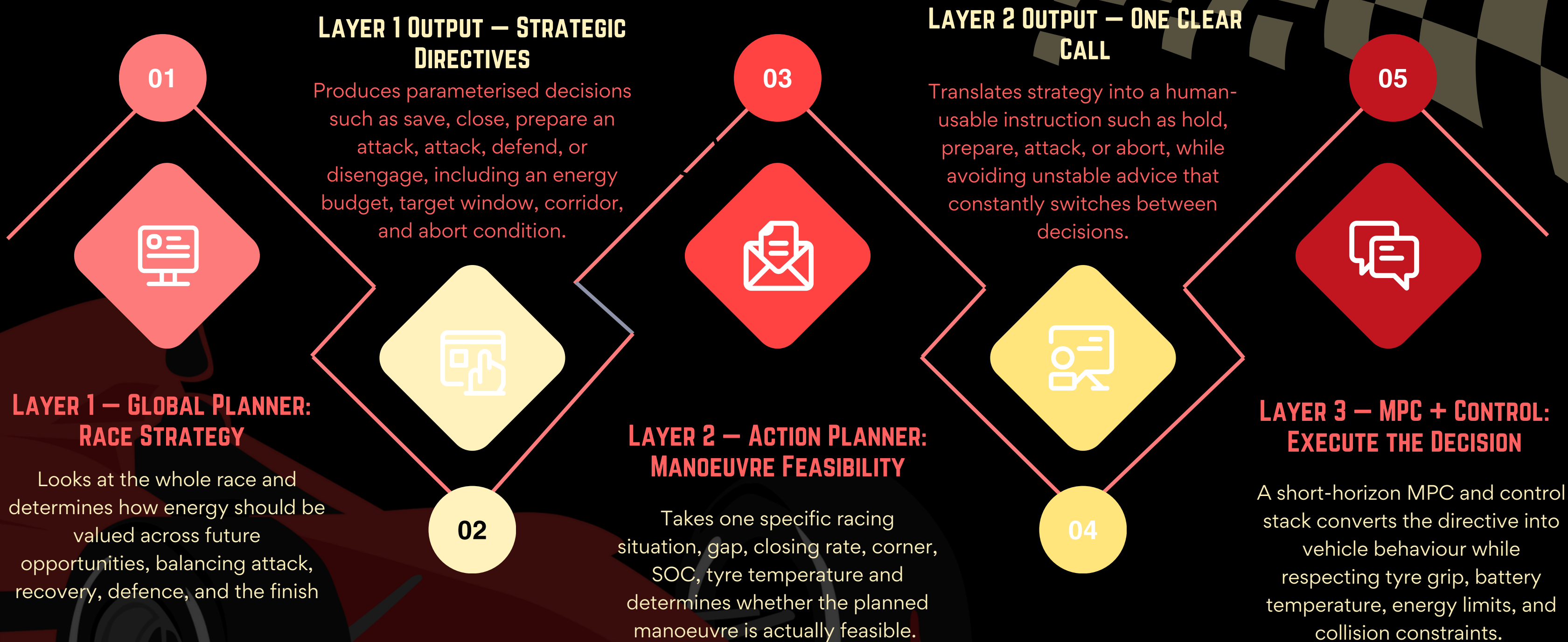
05

SUGGESTIVE, NOT AUTONOMOUS

The product gives the driver and pit wall actionable recommendations—hold, prepare, attack, defend, or abort—while keeping the human driver in control.



SOLUTION OVERVIEW



GLOBAL PLANNER WRITES THE RACE → ACTION PLANNER WRITES THE NEXT SCENE → MPC/CONTROL EXECUTES THE SCENE.

SIMULATION STACK

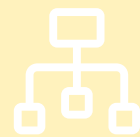
01



2D WEB SIMULATOR - PROVE THE INTELLIGENCE

Started with a lightweight Silverstone model focused on gap, closing rate, SOC, race position, and strategic decisions rather than visual realism.

02



WHAT WE LEARNED IN 2D

Established core strategic insights: energy has future value, following is not the same as racing, race distance must be unwrapped, and a pass must be retained to be valuable.

03



3D / TRACKMANIA-STYLE PHYSICS — VALIDATE THE MANOEUVRE

Moved from a point on a track spline to a 3D vehicle and environment to test left/right corridors, spatial positioning, launches, tows, and actual overtaking situations.

04



3D GAZEBO + ROS 2 — REALISTIC PHYSICS

Introduced a robotics-oriented plant with vehicle dynamics, contacts, inertia, sensors, and synchronised control, using C++ alongside Python

2D: IS THE STRATEGY RIGHT? → 3D: DOES THE MANOEUVRE LOOK REALISTIC? → GAZEBO: WOULD A VEHICLE UNDER REALISTIC PHYSICS ACCEPT IT?